SPA617 — RF Components: Characterization and Measurement

Experiment 2: Characterization of a Low-Pass Filter

Objective

To characterize the transmission response of a low-pass RF filter by measuring the output power at different input frequencies and determining its insertion loss and cutoff frequency.

Components Required

Low-pass RF filter, RF signal source, spectrum analyzer/VNA, RF cables and connectors, and suitable 50 Ω termination/load.

Theory

A low-pass filter (LPF) allows frequency components below its cutoff frequency to pass while attenuating components at higher frequencies. The insertion loss is calculated as:

IL (dB) = P_{in} − P_{out}

Here, P_{in} is the input/reference power measured without the filter and P_{out} is the output power measured after inserting the filter. The –3 dB cutoff frequency is measured relative to the minimum insertion loss in the passband.

Experimental Setup

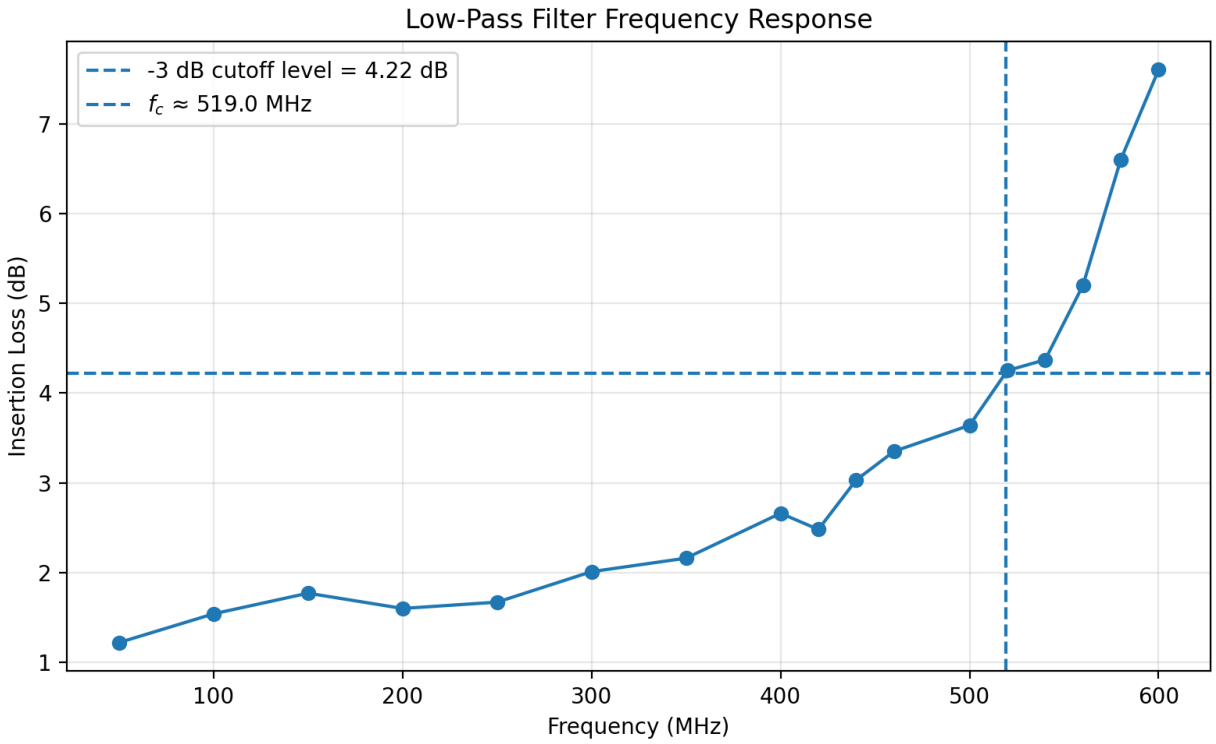
The RF source was connected to the measuring instrument through the low-pass filter. The reference power, P_{in}, was first measured without the filter. The filter was then inserted and the output power, P_{out}, was measured for different input frequencies. The input power was maintained at 0 dBm.

Observation & Graphs

Input Frequency (MHz)	P_in without filter (dBm)	P_out with filter (dBm)	IL (dB)
50	0.00	-1.22	1.22
100	0.00	-1.54	1.54
150	0.00	-1.77	1.77
200	0.00	-1.60	1.60
250	0.00	-1.67	1.67
300	0.00	-2.01	2.01
350	0.00	-2.16	2.16
400	0.00	-2.66	2.66
420	0.00	-2.48	2.48
440	0.00	-3.03	3.03
460	0.00	-3.35	3.35
500	0.00	-3.64	3.64
520	0.00	-4.25	4.25
540	0.00	-4.37	4.37

Input Frequency (MHz)	P_in without filter (dBm)	P_out with filter (dBm)	IL (dB)
560	0.00	-5.20	5.20
580	0.00	-6.60	6.60
600	0.00	-7.60	7.60

Frequency response of the measured low-pass filter:



Calculation & Results

The input/reference power was maintained at $\mathbf{P_{in} = 0\ dBm}$. Therefore, at each frequency:

$$\mathbf{IL = P_{in} - P_{out}}$$

For example, at 500 MHz, P_{in} = 0 dBm and P_{out} = –3.64 dBm. Therefore, IL = 0 – (–3.64) = **3.64 dB**.

The minimum measured insertion loss is 1.22 dB at 50 MHz. Hence, the –3 dB cutoff level relative to the minimum passband loss is 1.22 + 3 = **4.22 dB**.

At 500 MHz, IL = 3.64 dB and at 520 MHz, IL = 4.25 dB. Interpolation gives a cutoff frequency of approximately **519 MHz**, which can be reported as approximately **520 MHz**.

Result: The measured cutoff frequency of the low-pass filter is approximately **520 MHz**. The filter passes lower frequencies with comparatively low insertion loss and attenuates higher frequencies.

Conclusion / Key Takeaway

The transmission characteristics of the low-pass RF filter were successfully studied. The measured output power decreases as the frequency increases beyond the passband. The –3 dB cutoff occurs near 520 MHz, demonstrating the expected low-pass filtering behavior.

Note: P_{in} denotes power measured without the filter, while P_{out} denotes power measured with the filter connected.